

Digital control of bidirectional DC-DC converters in Smart Grids

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Abstract- The interest in the development towards intelligent microgrids which use renewable energy as primary sources has significantly increased in the last two years. The present paper examines and proposes a new digital control solution for bidirectional DC-DC converters. In order to control the converter in both directions the novel algorithm uses the current that charges and discharges the storage devices. The study treats and solves problems related to the battery charging algorithms and the power flow direction change.

I. INTRODUCTION

Recently, the interest in developing towards intelligent microgrids using renewable energy as primary sources has significantly increased. Although the intelligent microgrids are not new they are currently receiving increased attention because they use combined primary energy resources, improve power quality, and bring new services to the power grid (e.g. storage) [1]. A Smart Grid is a digital operating network that controls both interconnected loads and distributed energy resources, including renewable energy sources in such a manner that will allow the grid to operate efficiently with an existing power grid or in an independent mode. It can improve its power distribution efficiency according to the design of the grid and customer usage. A Smart Grid may incorporate 21st century technology but the term and the concept of “Smart Grid” stands not only for the latest technology used but also for the algorithm employed to control it. According to the United States Department of Energy’s Modern Grid Initiative report, the algorithm that controls the grid must be “self-healing”, secure, able to support widespread use of distributed generation, easy to manage and achieving high performance at a relative low production cost [2, 3].

Fig. 1 presents the topology of a standard smart micro-grid. As it can be seen in Fig. 1, a DC bus is needed (usually 400V) for supplying the AC bus through inverters, supplying DC loads and for charging the storage elements. For charging and discharging the storage devices, e.g. batteries and supercapacitors, bidirectional converters are used. A bidirectional converter is a DC-DC converter that allows power to flow in both directions.

Different control strategies for the bidirectional converter are presented in [4, 5, 6, 7, 8 and 9]. The control strategies focus on the converter working alone or in automotive applications. However none of these methods consider the

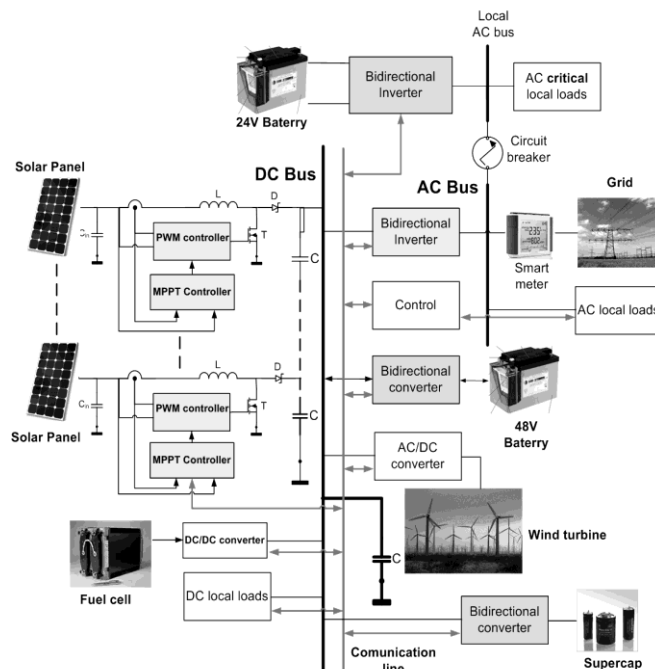


Fig. 1. Typical Smart-Grid topology

unitary functionality of a bidirectional converter and the storage devices in a microgrid. The control system together with the inverter, the bidirectional converter and the storage configuration will allow bidirectional power flow converting the microgrid into a current source or a current sink depending on the energy demand and generation.

The present paper examines new digital control solutions and proposes a new control method for bidirectional DC-DC converters. The paper will approach, discuss and attempt to solve some challenges inherent to Smart Grid bidirectional converters such as: current mode control in battery charging algorithms, anti-windup and power flow direction change. After implementing the topology of the converter and developing the control method, experimental results were obtained and were compared with the simulated ones. The article has 7 main sections: the first section presents notions related to the smart microgrid; section II treats the chosen topology design; section III treats algorithms for storage device charging; section IV proposes the control strategy; section V illustrates the simulation results; section VI presents the experimental results and the last section draws the conclusions.

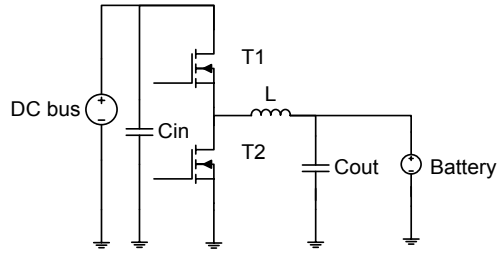


Fig. 2. Chosen bidirectional converter topology

II. TOPOLOGY DESIGN

From the previous section of this paper it is clear that power converters are an essential part of Smart Grids. Power converters enable electrical connection between renewable energy resources, storage and the grid. The development of these power converters with modular design, lower cost and improved reliability will improve the overall cost and durability of renewable energy systems integrated in Smart Grids.

The topology chosen to point out the advantages and disadvantages of the proposed control structure is a synchronous buck converter, illustrated in Fig. 2. This is the simplest topology that can achieve bidirectional power flow. In the direct power transfer mode, when the converter charges the battery, it behaves like a synchronous buck converter and in the reverse transfer mode, when the converter supplies the

DC bus from the battery it acts like a synchronous boost converter [10]. For low power applications the converter can be used like a standard buck or boost converter.

In Smart Grid applications the bidirectional converter must fulfill the following requirements: i) it should transfer energy from/to the storage device; ii) it should implement charging algorithms for different storage elements: Lead-acid, Ni-Cd, Li-ion Batteries or Supercapacitors; iii) it should be easily controllable; iv) it should be able to receive commands from a power management controller.

The design of the power topology should consider the behaviour of the converter both in the direct and reverse modes. The relationships used for the converter dimensioning are mentioned in the Table 1. In this table the following quantities represent: V_{in} – the input voltage of the converter in the considered mode; V_{out} – the output voltage of the converter in the considered mode; ΔV_{out} – the output voltage ripple; F_{sw} – the switching frequency; T_{sw} – the switching period; I_{sl} – the minimum current on the load when the current through the inductor is 0; I_{out} – the current through the load.

III. STORAGE DEVICE CHARGING AND DISCHARGING ALGORITHMS

In the direct transfer mode the control of the converter will employ a battery charging algorithm. The most popular algorithm for battery charging is constant current/constant voltage [13].

TABLE I
DESIGN RELATIONSHIPS

Mode	Buck	Boost
Duty cycle	$d = \frac{V_{out}}{V_{in}}$	$d = 1 - \frac{V_{in}}{V_{out}}$
Inductance	$L = \frac{V_{in}}{F_{sw} \cdot 2I_{sl}} \left(1 - \frac{V_{out}}{V_{in}} \right)$	$L = V_{out} \cdot d \frac{(1-d)^2}{F_{sw} \cdot 2I_{sl}}$
Output capacitor	$C_{out} = \frac{1}{8} \frac{\left(1 - \frac{V_{out}}{V_{in}} \right) V_{in} T_{sw}^2}{L \Delta V_{out}}$	$C_{out} = \frac{I_{out} d}{\Delta V_{out} F_{sw}}$
Maximum current through the transistor	$I_{tmax} = I_{out} + \frac{V_{out}}{F_{sw} L} \left(1 - \frac{V_{out}}{V_{in}} \right)$	$I_{tmax} = \frac{I_{out}}{1-d} + \frac{V_{out} d (1-d)}{2 F_{sw} L}$
Maximum voltage on the transistor	$V_{tmax} = V_{in}$	$V_{tmax} = V_{in}$
Maximum current through the diode	$I_{dmax} = I_{tmax}$	$I_{dmax} = I_{tmax}$
Maximum voltage on the diode	$V_{dmax} = V_{in}$	$V_{dmax} = V_{out}$

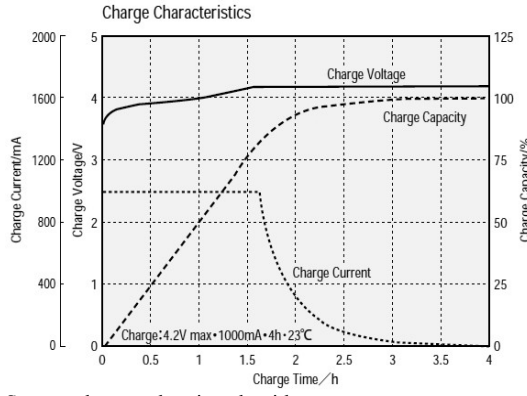


Fig. 3. Storage element charging algorithm

This charging algorithm can be used to charge almost all types of storage devices: Lead-acid batteries, Li-ion batteries or supercapacitors [14]. Only Ni-Cd batteries require a different end of charge detection usually based on battery voltage slope or temperature [11, 12].

The algorithm that will be implemented in the direct mode is described in Fig. 3. In the first state, when the battery is discharged the converter must supply a constant current. This current is maintained constant until the voltage on one storage cell reaches a certain limit (4.2V for a Li-ion cell for example) then the voltage is kept constant and the current decreases. At the end of the charging process a minimum current is supplied to the battery to compensate the self discharge phenomenon.

In the reverse mode, the converter must supply to the DC bus the amount of power dictated by the energy management system. During this time, the storage element is discharged. The discharging must be stopped when the storage device voltage drops under a predefined limit.

IV. CONTROL METHOD

The proposed bidirectional system that can fulfill the above mentioned features is presented in Fig. 4 a). The bidirectional converter is composed by: a voltage controlled bidirectional current source, I_{conv} , two PID controllers for the input and

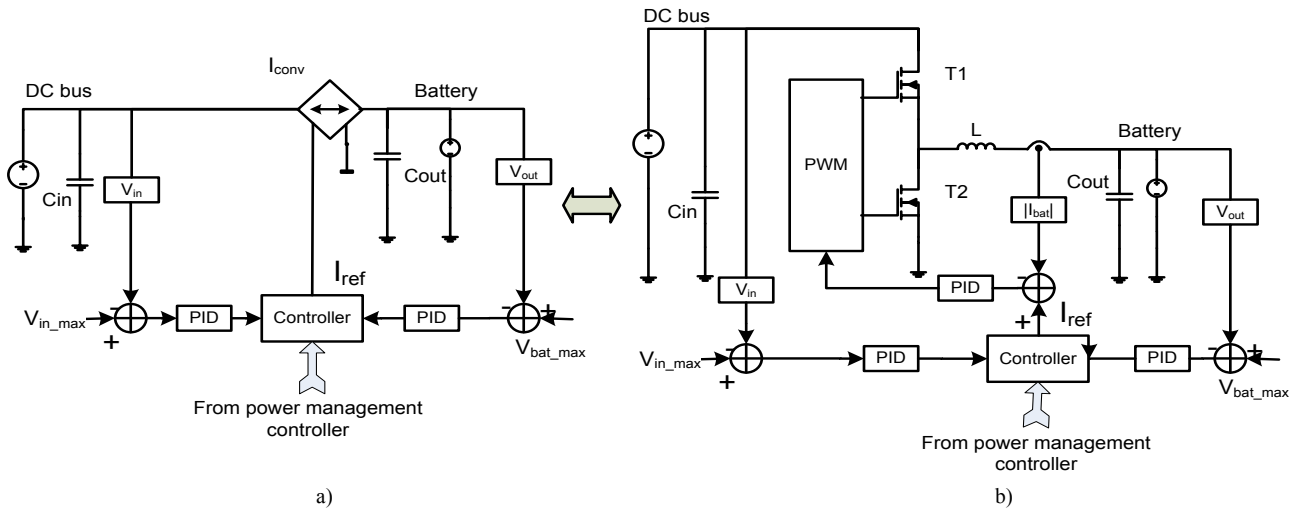


Fig. 4. Proposed control structure

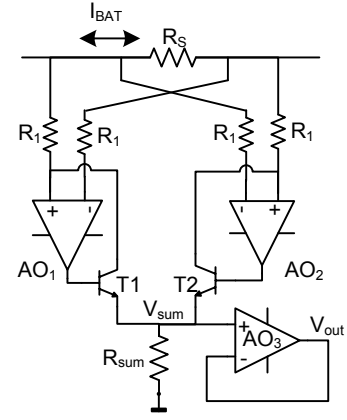


Fig. 5. Current sensor schematic

output voltage loops and a controller which decides the sense and the amount of power flow, based on data received from the power manager controller. The current source I_{conv} , is designed using the buck/boost converter described above and a current loop which keeps the current through the inductor constant (Fig. 4b). Because the battery is the one element from the Smart Grid that has a short life-time and a rather high price the current to and from it must be precisely controlled in order to extend its life time to the maximum.

The inductor current can be positive if the converter charges the battery, or negative if the converter supplies the DC bus because of the bidirectional nature of the converter. Besides current measurement, the current sensor must be designed also to select the absolute value of the inductor current. In this way the measured current will be positive regardless of the direction of the power flow.

The electrical circuit of the current sensor is presented in Fig.5. The sensor uses two standard current measurement circuits: a first one formed of AO_1 , T_1 , R_1 and a second one formed of AO_2 , T_2 , and R_1 . Each of these sensors measures the current in one direction, in the other direction the output being 0. The currents measured by these sensors are summed by R_{sum} . AO_3 is used as a voltage buffer.

The output voltage of the current sensor is:

$$V_{out} = \frac{|I_{BAT}| \cdot R_s \cdot R_{sum}}{R_1} \quad (1)$$

where I_{BAT} is the battery measured current, V_{out} is the measured value of the battery current R_1 , R_{sum} are the resistances presented in Fig. 5., and R_s is the current measurement resistor. R_1 is usually given by the producer of the current sensor and R_{sum} and R_s are chosen in such a way to obtain V_{out} equal to the full scale voltage, V_{FS} , of the ADC (analog to digital converter) at the maximum measured current. Whereas R_s is usually chosen based on power dissipation considerations (2), R_{sum} can be determined as in (3).

$$R_s = \frac{P_{R_s \max}}{I_{BAT \max}^2} \quad (2)$$

where $P_{R_s \max}$ is the maximum power dissipated on the measurement resistor and $I_{BAT \max}$ is the maximum designed battery current.

$$R_{sum} = \frac{R_1 \cdot V_{FS}}{|I_{BAT}| \cdot R_s} \quad (3)$$

The PID controller can be designed starting from an analog controller and digitize it using the Tustin transformation [14, 15 and 16]:

$$G_T(z) = \frac{[K_p \cdot (1 - z^{-2}) + K_d \cdot (z^{-1} - 1)^2 + K_i \cdot (z^{-1} + 1)^2]}{1 - z^{-2}} \quad (4)$$

Using mathematical manipulation of Eq. 4 one can obtain the new value of the sample $u[n]$ based on the previous value of the sample and the last two values of the error:

$$u[n] = u[n-1] + (Kp + Ki + Kd) \cdot e[n] - (Kp + 2Kd) \cdot e[n-1] + Kd \cdot e[n-2] \quad (5)$$

where Kp is the proportional gain, Ki the integral gain and Kd the derivative gain. The above relationship is implemented on a fix point DSC (digital signal controller) using Q24 format.

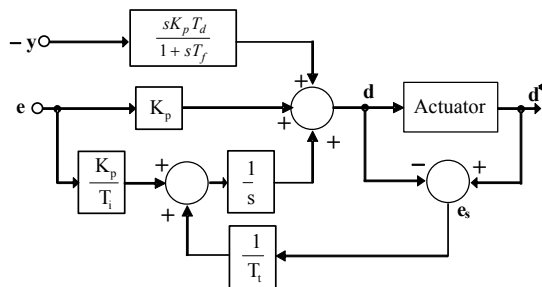


Fig. 6. Anti-windup algorithm

One important issue of this implementation that must be considered for this digital control topology is the windup effect. In the direct transfer mode, in the first charging cycle when the current is constant, the voltage PID controller is saturated to its maximum output value. During this time the integral element increases. If the calculations are implemented in a fixed point format this element can reach high values. When reaching the predefined threshold the voltage PID controller must come out of saturation to keep the battery voltage constant. This is impossible because of the high value of the integral element. This problem can be solved using an anti-windup algorithm. The PID algorithm presented above will be modified as in Fig. 6 to also implement the anti-windup.

When changing the direction of the power flow, because the current sensor computes the absolute value of the inductor current the measurement result will have the same value for two different currents. Because of this phenomenon there is an area that puts the converter in an unknown state when the energy direction is changed. Two methods for solving this issue are treated next.

One method implies using the converter as a synchronous converter in both transfer modes. After receiving the command for switching the direction, the converter will be controlled with a constant duty cycle that will change the direction of the current and then the PID algorithms will be applied for further converter control. The time necessary for the direction change is also calculated by the algorithm. The next chart illustrates this statement (Fig. 7).

When the changing of direction command is received a fixed duty cycle D_{fix} is chosen - needed to change the direction of the inductor current. The choice is made based on considering the duty cycle in the steady state operation and the input and output voltages. After D_{fix} is established the time, ΔT , necessary to change direction of the power flow can be evaluated in software with the following relationship:

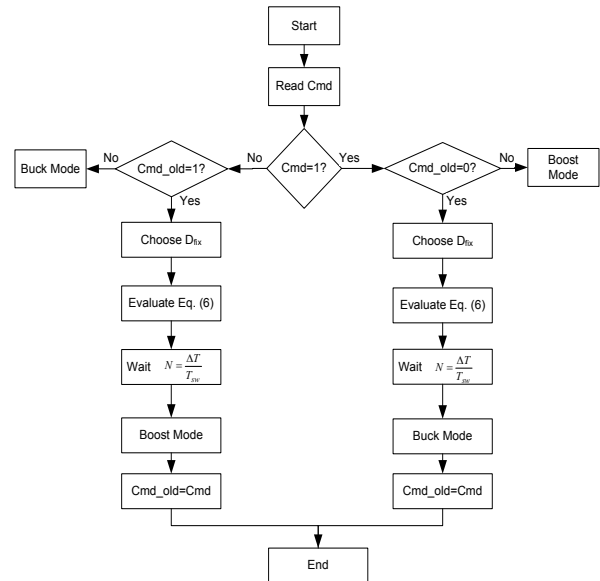


Fig. 7. Software implemented diagram

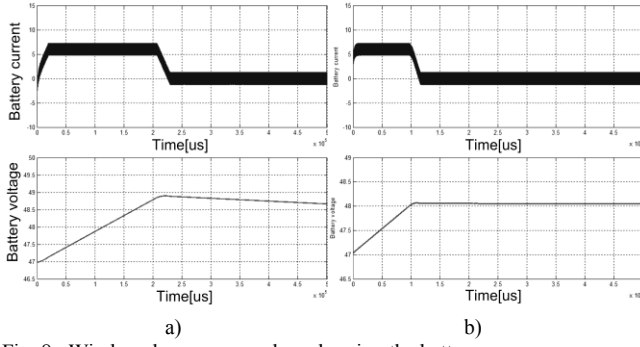


Fig. 8. Windup phenomenon when charging the battery

$$\Delta T = L \cdot \frac{\Delta \hat{i}_L}{V_{in} \cdot D_{fix} - V_{out}} \quad (6)$$

where $\Delta \hat{i}_L$ is the excursion of the average inductor current from its positive to its negative value, L is the inductor value, V_{in} the input voltage and V_{out} the output voltage. The algorithm then applies D_{fix} , waits a number N cycles and applies the PID algorithms suited for the changed direction. The number of cycles is calculated as follows:

$$N = \frac{\Delta T}{T_{sw}} \quad (7)$$

where T_{sw} is the switching periode.

A particular case of the method presented above implies using the converter as a standard buck converter in direct transfer mode and using it as a standard boost converter in reverse transfer mode. The internal diodes of the transistors will be used for achieving this purpose. This method can be used in low power applications because of its low efficiency.

V. SIMULATION RESULTS

The above simulation model of the bidirectional converter is implemented in Matlab 2011b and represented in Fig. 9. The model has five blocks: the power converter, the PWM generator, the constant voltage bus, the storage device and the control block. The control block will contain the algorithm

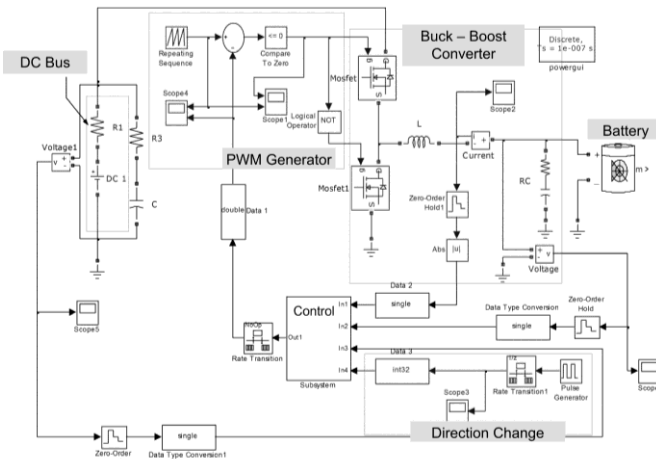


Fig. 9. Simulation model of the bidirectional converter

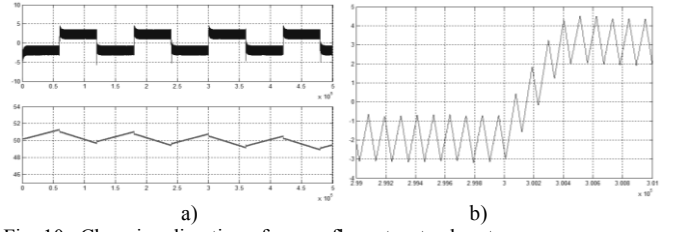


Fig. 10. Changing direction of power flow at a steady rate

for switching the direction and PID algorithm for controlling the converter.

The implementation of the control algorithms without anti-windup is presented in the Fig. 8a).

The results of implementing the algorithms with anti windup are represented in Fig 8b). It can be stated that, because of the anti-windup loop, the voltage on the battery does not exceed 48V from the moment it reaches this value.

The simulation results for the implemented changing direction algorithm are presented in the Fig.10a). The direction is change with a steady rate to illustrate the behavior of the converter. Usually the command for the direction change comes from the energy management controller. The upper trace of Fig 10a) illustrates the current charging and discharging the supercapacitor, and the lower one the voltage over the storage device.

A zoom in the change of the current direction is represented in the Fig. 10b). The value calculated for the time of the direction change is $\Delta T = 32\mu s$ and the number of cycles is $N = 3.2$ steps

VI. EXPERIMENTAL RESULTS

For testing the proposed control method, the setup in Fig 11a) is used. A supercapacitor of 10F/2.5 V will be the storage device and the DC current bus consists of a power supply in constant current mode connected in parallel with an electronic load functioning in constant voltage mode. For practical implementation Matlab/Simulink was used. The code was automatically generated and downloaded in a DSC (digital signal controller) from Texas Instruments, C2000 family, TMS320F28027 Piccolo.

Figure 12 presents the results when the converter works with the anti-windup control loop and without it. After the first 15 seconds the voltage on the battery should remain constant but because the integral component of the PID algorithm cumulates error the PID algorithm does not come out of the saturation region until this component returns at the prescribed value, Fig. 12a).

It can be observed that the time it takes the algorithm to get

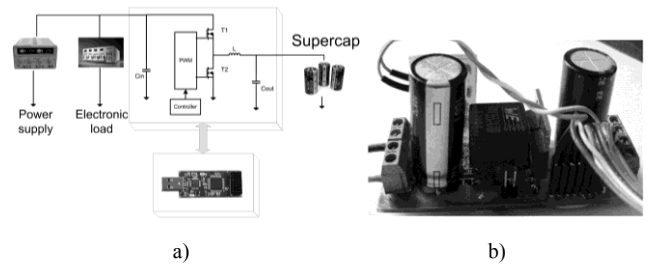


Fig. 11. Converter implementation

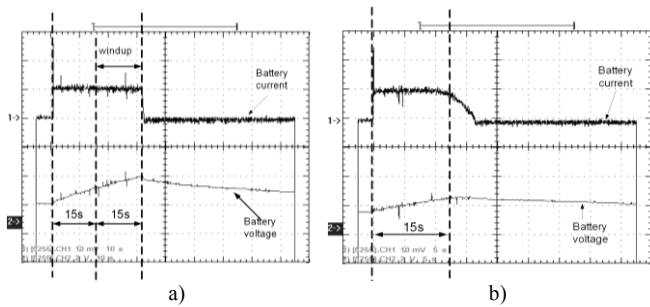


Fig. 12 Converter without anti-windup

out of saturation is approximately equal to the time the current is constant. This time can reach several hours, if batteries are used as storage devices. This effect is avoided using the anti-windup algorithm presented above and the result is shown in Fig 12b). Compared to a digital PID controller, an analog controller will saturate at the supply voltage of the operational amplifier that implements the PID, so the anti-windup is implemented implicitly in analog controllers.

The behavior of the converter when the power flow direction is changed is illustrated in Fig 13. The upper trace illustrates the current charging and discharging the supercapacitor, and the lower one the voltage over the storage device.

VII. CONCLUSIONS

The aim of the present paper was to point out and solve problems regarding the connection of a bidirectional converter in a Smart Grid application. A control algorithm suited for bidirectional converters used in intelligent microgrids was proposed. For the control of the converter in both directions, this algorithm uses the current that charges and discharges the storage devices. The algorithm proposed can be used with any kind of storage device: Li-ion, lead-acid batteries or supercapacitors. The study revealed and solved problems regarding the battery charging algorithms and the change of the power flow direction. A current sensor that implements also the absolute value of the current was developed to ease the implementation of the current ridings. An improved battery charging algorithm with anti-windup was analyzed and tested. Furthermore two algorithms for the direction change were implemented and tested. It can be observed that the experimental results are similar to the simulated ones. The converter is illustrated in Fig 11b).

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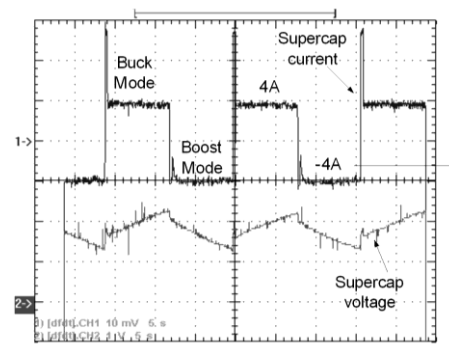


Fig. 13. Results of the change of direction method

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